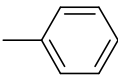


**Electronic effects:**

|                                                                                                                                                                                                                                    | Electron-withdrawing groups<br>(they “take” electron density<br>away from groups they are<br>attached to)                                                                                                                                                                                                                      | Electron-donating groups<br>(they increase electron<br>density of groups they are<br>attached to)                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inductive effect (electronegativity)                                                                                                                                                                                               | <p>-F, -Cl, -Br, -I</p> <p>-OH, OR</p> <p><math>^{-+}\text{OH}_2</math>, <math>^{-+}\text{OHR}</math></p> <p>-NH<sub>2</sub>, -NHR, NR<sub>2</sub></p> <p><math>^{-+}\text{NH}_3</math>, <math>^{-+}\text{NH}_2\text{R}</math>, <math>^{-+}\text{NR}_2\text{H}</math>,</p> <p><math>^{-+}\text{NR}_3</math></p> <p>-C=O(R)</p> | <p>Alkyl groups (such as -CH<sub>3</sub>,<br/>-CH<sub>2</sub>R, -CHR<sub>2</sub>, -CR<sub>3</sub>, etc)</p>                                                       |
| Resonance effect (hyperconjugation<br><u>not</u> considered here) (Resonance<br>forms are possible)                                                                                                                                | -C=O(R)                                                                                                                                                                                                                                                                                                                        | <p>-OH, OR, -NH<sub>2</sub>, -NHR,<br/>NR<sub>2</sub></p> <p>-C=C, -C≡C, </p> |
| <p>If a group (such as -OH) has opposite effects through inductive and resonance effect, the resonance effect wins (<b>IF</b> resonance is possible). If resonance is not possible, then only the inductive effect is present.</p> |                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                   |

### **Basic vs nucleophilic properties of functional groups:**

In Organic Chemistry, we differentiate between a “base” and a “nucleophile”. In reality, both type of compounds are Lewis bases (they need a pair of electrons that can be used to form a new bond with another compound).

In Organic Chemistry, when we say “base”, we mean a Lewis base that will form a new bond with H ( $H^+$ , which we either write as “ $H^+$ ” OR, as HX-the bond between H and X will be broken as H is attacked by the base).

In Organic Chemistry, when we say “nucleophile”, we mean a Lewis base that will form a new bond with C ( $C^+$ , or  $C\delta^+$ ).

Nucleophile, is also a term relating kinetics: a stronger nucleophile will react faster. While base, is usually a thermodynamic term: a stronger base will react to a larger extent (in a possible equilibrium).

### **Comparing basic and nucleophilic strength:**

It is important to remember that this is a complex topic. THEREFORE, we need to simplify for this course. To keep it simple we will compare base/nucleophile strength on systems which are VERY similar with only one structural variable different: FOR example:  $H_2O$  vs  $^-OH$  (everything the same, but one is charged, the other isn't). OR  $^-F$  vs  $^-Cl$  (same valence electrons, but different period). OR  $NH_3$  vs  $H_2O$  (same period, same atom attached to central atom, same charge, but different family or periodic table group). It isn't easy (therefore we will not do it) to compare bases/nucleophiles with more than two structural changes: for example  $^-OH$  vs  $NH_3$ .

Remember acid/base Bronsted theory (each acid will have a conjugate base, and viceversa): the stronger the acid, the weaker the conjugate base (the opposite is also true):

SO, HCl is a strong acid ( $^-Cl$  is a weak base).  $H_2O$  is a weak acid ( $^-OH$  is a strong base).

**Normally**, the stronger the base, the stronger the nucleophile. (See exceptions below).

So, for example:

-If everything else is the same, the negatively charged species is always a MUCH stronger base and nucleophile:

$^-OH > H_2O$  ;  $^-OR > HOR$  ;  $^-NH_2 > NH_3$  ;  $^-NR_2 > NR_3$  ; etc

-If everything else is the same, if you add an electron donating group by inductive effect (e.g. alkyl group) to the main atom, the stronger the base AND nucleophile:

$^-OR > ^-OH$  ;  $HOR > H_2O$  ;  $^-NR_2 > ^-NH_2$  ; etc

-If everything else is the same, going left in the same period of the periodic table increases the strength of the base AND nucleophile:

$^-CH_3 > ^-NH_2 > ^-OH > ^-F$  ;  $NH_3 > H_2O$  ( $CH_4$  and  $HF$  are NOT Lewis bases) ; etc

Main exception to the rule “the stronger the base, the stronger the nucleophile”: If everything else is the same, going down in the same group increases the strength of the nucleophile, BUT, decreases the strength of the base: SO:

Base trend:  $\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$  ; Nucleophile trend:  $\text{F}^- < \text{Cl}^- < \text{Br}^- < \text{I}^-$

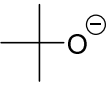
Base trend:  $\text{H}_2\text{O} > \text{H}_2\text{S}$  ;  $\text{HOR} > \text{HSR}$  ;  $\text{OH}^- > \text{SH}^-$  ;  $\text{OR}^- > \text{SR}^-$  ;

Nucleophile trend:  $\text{H}_2\text{O} < \text{H}_2\text{S}$  ;  $\text{HOR} < \text{HSR}$  ;  $\text{OH}^- < \text{SH}^-$  ;  $\text{OR}^- < \text{SR}^-$

Base trend:  $\text{NH}_3 > \text{PH}_3$  ; Nucleophile trend:  $\text{NH}_3 < \text{PH}_3$

If you remember only a few examples, please remember:

Good/OK nucleophiles but weak base:  $\text{Cl}^-$  ,  $\text{Br}^-$  ,  $\text{I}^-$

Strong bases poor nucleophiles:  $\text{NaH}$ ,  $\text{LiH}$ ,  $\text{KH}$ , 

Good nucleophile, good base:  $\text{OH}^-$

Poor nucleophile, poor base:  $\text{H}_2\text{O}$ ,  $\text{ROH}$ .

Substitution vs Elimination:

**$\text{S}_\text{N}2$ :** Favoured by: -Primary carbon, good nucleophile (but possibly not too strong a base or  $\text{E}2$  will compete), solvent not clear (but acetone is a good example).

**$\text{S}_\text{N}1$ :** Favoured by: -Tertiary carbon, NON-basic weak/not too strong nucleophile, polar solvent. Carbocation will NOT form in basic medium.

**$\text{E}2$ :** Requires H on carbon next to carbon with Leaving Group and strong base (but possibly not too nucleophilic or  $\text{S}_\text{N}2$  will compete). Leaving group can be on any type of carbon (because the base will take any hydrogen on a carbon next to it).

**$\text{E}1$ :** Requires tertiary carbon, polar solvent, non-nucleophilic very weak Lewis base, acidic/neutral medium (Carbocation will NOT form in basic medium).

**Suggestions on how to work with mechanisms:**

**1- From the reagents: Identify:**

- a. Type of reaction
- b. Product if possible

**THEN, before you write any mechanism, go back to the reagents AND:**

**2- IDENTIFY:**

- a. **FIRST** Acids and bases (Organic Chemistry definition) and use curly arrows to draw their reactions, and write the product structure of this step, and,
- b. **ONLY THEN**, Nucleophiles and electrophiles, and use curly arrows to draw their reaction, and write the product structure of this step, **THEN** repeat a/b loop until all groups have been used
- c. You might need a final protonation/deprotonation step to get to the final product, **ONLY THEN** proceed to 3

**3- Go back to 1 and make sure that what you wrote in 2 makes sense with what you thought in 1. If it doesn't, then either is wrong. Please go back and check.**

**4- At every step: make sure you write the produce of each step, not following your knowledge of the mechanism, but following what the curly arrows you have used indicate: if they don't coincide with your knowledge of the mechanism, then this means is likely your curly arrows are wrong (OR your knowledge of the mechanism is wrong)**

**5- If any of the intermediates is stabilised by resonance, please write all the resonance forms that are relevant for full marks.**